

Work Assignment No. 3	
Course	Data Modelling in Engineering
Academic Yea	2025/2026
Objective	Knowledge Modelling using Frames and UML
Classes	2 classes x 3 classes + 3 extra hours
Submission Date	2026/06/02
Specific Objectives: 1. Frame-based Modelling a. Understand the concept of knowledge modelling using Frames and its relevance in the context of Electrical and Computer Engineering. b. Represent models using the Prolog language and the provided GOLOG library. c. Model information using concepts such as: frames, slots, relations, inheritance mechanisms, methods and demons. 2. UML Modelling a. Model the entities of the system and their relationships using: Class Diagrams and Sequence Diagrams.	

Assignment Objectives

This assignment aims to provide students with practical contact with the concepts of knowledge modelling using Frames supported by the Prolog language and the GOLOG library, as well as UML modelling focused on the structural and behavioural representation of intelligent supervision and automation systems within the context of Electrical and Computer Engineering.

The assignment is divided into two parts:

Part I – Frame-based Modelling,

Part II – UML-based Modelling.

Problem Description

In this project, students must model and implement a **Smart Warehouse Management System**.



Figura 1 - Smart Warehouse (AI-generated image created with ChatGPT (OpenAI), 2026, adapted by the authors)

The system shall integrate: environmental sensors; automatic climate control; stock management; automatic alarms; order management; dynamic adaptation of warehouse conditions.

The warehouse shall include sensors responsible for monitoring the: temperature; humidity; air quality; presence/occupancy; and door status; and also actuators responsible for: cooling; ventilation; humidification; alarms; and automatic door control.

System Functionalities

The system shall allow:

- administrators to create and configure smart warehouses;
- administrators to configure sensors and actuators;
- administrators to parameterize climate levels;
- employees to manage products and stock;
- employees to update information regarding stored products;
- customers to place orders containing different products;
- customers to consult on the status of their orders;
- the system to automatically generate alarms and control climate mechanisms.

Main System Elements

The following entities shall be considered:

- **warehouse** characterized by name; location; capacity; zones; current temperature; operational state.
- **product** characterized by name; reference; category; quantity; expiration date; ideal temperature; warehouse location; unit price.
- **order** characterized by reference; included products; quantity; price; order state.
Possible order states: pending preparing shipped delivered
- **sensor** characterized by identification; type; current value; location; state. Possible types: sensor_temp sensor_humidity sensor_co2 sensor_presence sensor_door
- **actuator** characterized by identification; type; state. Possible types: cooling_system ventilation humidifier alarm_system automatic_doors.

Stock and Order Management

Whenever a customer places an order, the system shall:

1. verify stock availability;
2. automatically update existing quantities;
3. prevent impossible orders;
4. generate appropriate warning messages in case of unavailability.

Intelligent Climate Control System

The warehouse climate system shall operate automatically and adaptively considering: environmental conditions; warehouse occupancy; warehouse operational state. For that, the system shall support five climate classifications: *freezing*; *cold*; *comfort*; *hot*; *burning*

The climate system shall operate automatically based on: sensor values; configured limits; number of occupants.

Climate levels may be parameterized using: upper limit (ls); absolute upper limit (las); lower limit (li); absolute lower limit (lai).



Figura 2 - níveis de climatização do sistema de climatização.

The climate system may assume three possible states: *inactive*; *warming* and *cooling* that are activated in result of the operations: stop, warm and cool. Additionally, the current temperature of the climate control system is provided by a thermostat. The control of this climate system is considered automatic and depends on the thermostat temperature.

Alarm System

There is also a subsystem that, whenever a critical situation occurs, automatically generates alarm messages whenever the absolute upper or absolute lower temperature limits are exceeded. Alarm messages are characterized by:

- **Event:** event that triggered the alarm message
- **Temp:** temperature measured at the moment of the alarm event
- **Date:** date and time of the alarm event

Occupancy Adaptation

Using a presence sensor, the system shall monitor warehouse occupancy and automatically adapt climate levels according to the number of occupants.

Thus, if the number of people in the space exceeds $X+Y$, the climate levels should decrease by $Z\text{ }^{\circ}\text{C}$; alternatively, if the number of people falls below $X-Y$, the climate levels should increase by $Z\text{ }^{\circ}\text{C}$.

Students may consider step-based increases/decreases or implement alternative adaptive solutions. Whenever an automatic variation of the climate levels occurs, an alarm message must also be generated.

User Interface

For interaction with the developed functionalities of the Warehouse Management System, an appropriate menu/interface must be implemented to support user interaction and allow:

- user interaction;
- system simulation;
- information consultation;
- execution of the main functionalities.

Part I – Frame-based Modelling

The study and implementation of the problem proposed in this part of the assignment aim to introduce the following concepts:

-  Problem analysis;
-  Knowledge structuring using Frames;
-  Relations and inheritance mechanisms;
-  Logic programming;
-  Implementation of methods and demons.

Modelling / Implementation

This problem shall be modelled through a Frame-based structure, using generic frames, concrete frames, relations, and inheritance mechanisms, in the most appropriate way possible. Note that some of the required functionalities will be better implemented using methods, while others will be better implemented using demons.

During the modelling of this problem, the following project phases must be considered:

1. Model the problem using Frames and implement it in Prolog using the provided GOLOG library.
2. Identify and implement the frames, slots, relations, and rules/predicates required to solve the proposed problem.
3. Implement the necessary methods and demons.
4. Develop a menu to simulate the system's operation.
5. Simulate the system's operation.

Part II – UML-based Modelling

In the UML component, students shall develop a model focused on the structural and behavioural representation of the system. For this purpose, a **Class Diagram** must be created representing: warehouse, products, sensors, actuators, orders, alarms, and relationships between entities. Additionally, a **Sequence Diagram** must be created representing: sensor readings, demon activation, actuator actions, and alarm generation.

Reflection Topics

This section consists of more reflective considerations that may serve as “inspiration” for completing the report/form. By the end of the assignment, students should be able to answer the following questions:

- Advantages/disadvantages of Frame-based modelling;
- How demons operate;
- Characteristics of UML modelling.

Functional Requirements

The solution to this problem requires the fulfilment of the following functional requirements:

Functional. Req.	Description
FR1	Create, visualize, update and delete warehouses
FR2	Create and manage sensors and actuators (including real-time simulation and updates)
FR3	Create, visualize and update products and stock
FR4	Create orders considering stock availability
FR5	Visualize order states.
FR6	Configure and simulate automatic climate control
FR7	Implement occupancy-based adaptive climate control
FR8	Generate and visualize alarm messages
FR9	Develop the supporting Class Diagram
FR10	Develop the Sequence Diagram illustrating sensors, demons, actuators and alarms

Mandatory Rules for Moodle Mini-Test Alignment

To ensure that students can correctly answer the Moodle mini-test, the following minimum functionalities must be properly implemented:

1. **Smart Warehouse Project:**

- a. Create frames for the following sensors: **sensor_temp**, **sensor_humidity**, and **sensor_CO2**.
- b. Create an **actuators** frame containing **cooling_system**, **ventilation**, and **humidifier**.
- c. Implement demons for automatic actuator control, for example: activating the cooling system whenever the temperature exceeds a predefined threshold.
- d. Implement occupancy-based adaptive logic.
- e. Generate automatic alarm messages.
- f. Implement simulation using **set_value** and **get_value**.

2. **UML:**

- a. a. Create a Class Diagram including frames, slots, and relationships.
- b. b. Create a Sequence Diagram illustrating sensors, demons, actuators, and alarms.

These functionalities will be validated in the Moodle test; therefore, students are required to implement and test them correctly.

Class Planning

This assignment should follow the plan below:

1st class:

- Presentation of the assignment and the tools to be used.
- Review of concepts related to Prolog programming using the GOLOG library.
- Initial system structure.
- Implementation of the intelligent climate control system.

2nd class:

- Implementation of the remaining functional requirements.
- UML diagrams.
- Finalization of the assignment.

Submission (Moodle)

Students must submit the following files through Moodle:

1. zip/rar file containing Part I (Prolog files including the menu/interface and simulation examples).
2. Demo video with the previous simulation examples (covering the implemented FR – FR1/FR8).
3. PDF containing the implementation of Part II (Class Diagram and Sequence Diagram).

Students must also complete the Moodle mini-test.

IMPORTANT: The Moodle mini-test will be directly based on the mandatory functionalities described in this assignment. Only students who correctly implement these functionalities will be able to adequately answer the mini-test.

Submissions sent by email will NOT be considered.

Assessment

Critérios de avaliação

Itens	Descrição	Cotação / penalização
Critérios	Air conditioning project implementation (Class 1 exercise)	3
	Smart Warehouse implementation	3
	Demonstration video	2
	Class Diagram	2
	Sequence Diagram	2
	Moodle mini-test	8
Total		20.0
Failure to submit one of the required items		-20.0

Theoretical-Practical Classes

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